

Beaver Brigade Radio Guide

Trail radio setup & privacy codes

FRS and GMRS basics for our group



EDUCATIONAL SECTION

Frequencies and licensing

462 MHz and 467 MHz

- FRS and GMRS share 22 numbered channels in these UHF bands.
- Direct communication uses the same transmit and receive frequency.

FRS vs. GMRS

- FRS radios require no individual license.
- GMRS operation requires a GMRS license, including direct radio-to-radio use.
- A license covers eligible family members. It does not cover unrelated club friends.

Channels and power limits

FRS handhelds can use 1–22. GMRS vehicle mobiles cannot transmit on 8–14.

For a group with vehicle mobiles, choose a shared channel in 1–7 or 15–22.

Channels	Band	FRS maximum	GMRS maximum
1–7	462 MHz	2 W ERP	5 W ERP
8–14	467 MHz	0.5 W ERP	0.5 W ERP, handheld only
15–22	462 MHz	2 W ERP	50 W output, vehicle mobile

ERP means Effective Radiated Power. It accounts for the radio's transmitter power, cable losses, and antenna gain relative to a standard dipole antenna.

For example, **5 watts from the radio with 3 dB of antenna gain** produces about **10 watts ERP**, assuming no cable loss. Basically it describes how effectively the antenna system sends the signal in its strongest direction.

ERP includes antenna gain. Actual radio output may be lower.

Range and antenna basics

Terrain controls the path

- UHF works best with a clear line of sight. Hills, canyon walls and vehicles can block it.
- There is no universal 3.5-mile limit. Antenna height at both ends matters.
- Higher, clearer antenna placement often helps more than extra watts.

Vertical antennas and gain

- Keep handheld and vehicle antennas upright. Different orientations weaken the link.
- An extra 3 dB of gain roughly doubles effective radiated power in the favored direction.
- Antenna gain does not increase the radio's transmitter wattage or guarantee twice the range.

Wide and narrow bandwidth

GMRS can use either

- GMRS allows wider voice signals on channels 1–7 and 15–22.
- Channels 8–14 require narrow bandwidth. FRS is narrow on every channel.
- A narrow transmission can sound quiet on a wide receiver. A wide signal may distort on a narrow receiver.

For this mixed group

- Use narrow bandwidth for the shared direct-channel exercise.
- MXT275 and GXT1000 use narrow. The Motorola T4xx is narrow.
- MXT575: menu bs, then bn.
- GM-15 Pro: Bandwidth / WN, then Narrow.

What privacy codes actually do

A filter for your speaker

- CTCSS uses a low-frequency tone. DCS uses a digital code.
- Your radio unmutes when the incoming signal matches its receive setting.
- Anyone listening with codes OFF can still hear you. Codes provide no encryption.

The channel is still shared

- Other users can still interfere with you. A quiet speaker does not prove a clear channel.
- Use Monitor or turn receive filtering off to listen before transmitting.
- Match the actual CTCSS frequency or DCS value and polarity. Manufacturer code numbers can differ.

HOW-TO SECTION

Beaver Brigade channel plan

Save both setups before the ride. The different channel provides the backup.

PRIMARY: channel 16

462.5750 MHz

CTCSS 67.0 Hz

Midland / Motorola: privacy code 1

Baofeng: TX and RX CTCSS 67.0 Hz

Start here for the group ride.

BACKUP: channel 17

462.6000 MHz

CTCSS 67.0 Hz

Midland / Motorola: privacy code 1

Baofeng: TX and RX CTCSS 67.0 Hz

Switch together when the lead calls it.

Both setups use direct mode and narrow bandwidth. Listen first: these are shared channels.

Midland MXT275

Use the standard direct channels. This model uses narrow bandwidth.

PRIMARY: channel 16, Ct 1

1. Use the microphone CHANNEL buttons to select 16.
2. Press MENU. Use VOLUME +/- to find Pt. Press LOCK.
3. Select Ct (CTCSS). Press LOCK.
4. Choose 1 (67.0 Hz). Press LOCK to save.
5. Leave the menu and recheck channel 16 with CTCSS code 1.

BACKUP: channel 17, Ct 1

1. Use the microphone CHANNEL buttons to select 17.
2. Press MENU. Use VOLUME +/- to find Pt. Press LOCK.
3. Select Ct (CTCSS). Press LOCK.
4. Choose 1 (67.0 Hz). Press LOCK to save.
5. Leave the menu and recheck channel 17 with CTCSS code 1.

After saving both, return to channel 16. Switch to 17 only when the group changes channels.

Midland MXT575

CTCSS code 1 = 67.0 Hz. Save the privacy setting separately on each channel.

PRIMARY: channel 16, Ct 1

1. Use the microphone CHANNEL buttons to select 16.
2. Press MENU. Use VOLUME +/- to find Pt. Press LOCK.
3. Select Ct (CTCSS). Press LOCK.
4. Choose 1 (67.0 Hz). Press LOCK to save.
5. Leave the menu and recheck channel 16 with CTCSS code 1.

BACKUP: channel 17, Ct 1

1. Use the microphone CHANNEL buttons to select 17.
2. Press MENU. Use VOLUME +/- to find Pt. Press LOCK.
3. Select Ct (CTCSS). Press LOCK.
4. Choose 1 (67.0 Hz). Press LOCK to save.
5. Leave the menu and recheck channel 17 with CTCSS code 1.

For each channel: MENU, bs, LOCK, bn (narrow), LOCK. Finish on primary channel 16.

Midland GXT1000VP4

First: MENU to Gr, select oF with ▲/▼, then PTT. Keep Group Mode off.

PRIMARY: channel 16, code 1

1. Press MENU to the channel setting. Use ▲/▼ to select 16. Press PTT.
2. Press MENU until the Off / CTCSS / DCS setting appears.
3. Use ▲/▼ to choose CTCSS. Press MENU.
4. Choose code 1 with ▲/▼. Press PTT to save.
5. Recheck channel 16, CTCSS and code 1.

BACKUP: channel 17, code 1

1. Press MENU to the channel setting. Use ▲/▼ to select 17. Press PTT.
2. Press MENU until the Off / CTCSS / DCS setting appears.
3. Use ▲/▼ to choose CTCSS. Press MENU.
4. Choose code 1 with ▲/▼. Press PTT to save.
5. Recheck channel 17, CTCSS and code 1.

After saving both, return to channel 16. These instructions use standard channels 1–22.

Midland GXT1000AZ

First: MENU to Grp, select oF with ▲/▼, then PTT. Keep Group Mode off.

PRIMARY: channel 16, code 1

1. Press MENU to the channel setting. Use ▲/▼ to select 16. Press PTT.
2. Press MENU until the Off / CTCSS / DCS setting appears.
3. Use ▲/▼ to choose CTCSS. Press MENU.
4. Choose code 1 with ▲/▼. Press PTT to save.
5. Recheck channel 16, CTCSS and code 1.

BACKUP: channel 17, code 1

1. Press MENU to the channel setting. Use ▲/▼ to select 17. Press PTT.
2. Press MENU until the Off / CTCSS / DCS setting appears.
3. Use ▲/▼ to choose CTCSS. Press MENU.
4. Choose code 1 with ▲/▼. Press PTT to save.
5. Recheck channel 17, CTCSS and code 1.

After saving both, return to channel 16. These instructions use standard channels 1–22.

Baofeng GM-15 Pro

Enter 67.0 Hz for both TX and RX. Follow menu names if firmware uses different numbers.

PRIMARY: channel 16, 67.0 Hz

1. Select the A/B line. Choose GMRS channel 16.
2. MENU to Tx CTCSS (12). MENU, choose 67.0 Hz with ▲/▼, MENU to save.
3. Set Rx CTCSS (10) to 67.0 Hz the same way.
4. Set Tx DCS and Rx DCS to OFF. Set Bandwidth / WN to Narrow.
5. EXIT. Recheck channel 16's saved tones.

BACKUP: channel 17, 67.0 Hz

1. Select the A/B line. Choose GMRS channel 17.
2. MENU to Tx CTCSS (12). MENU, choose 67.0 Hz with ▲/▼, MENU to save.
3. Set Rx CTCSS (10) to 67.0 Hz the same way.
4. Set Tx DCS and Rx DCS to OFF. Set Bandwidth / WN to Narrow.
5. EXIT. Recheck channel 17's saved tones.

If the tones will not save, use matching GM-15 Pro software. See speaker notes for the fallback.

Motorola T461

Use two-way radio mode. Code 1 = CTCSS 67.0 Hz. The T4xx family uses narrow bandwidth.

PRIMARY: channel 16, code 1

1. Press MENU/LOCK until the channel number flashes.
2. Use +/- to choose 16. Press PTT to save.
3. Press MENU/LOCK until the privacy-code number flashes.
4. Use +/- to choose 1. Press PTT to save.
5. Confirm channel 16 and code 1 on the display.

BACKUP: channel 17, code 1

1. Press MENU/LOCK until the channel number flashes.
2. Use +/- to choose 17. Press PTT to save.
3. Press MENU/LOCK until the privacy-code number flashes.
4. Use +/- to choose 1. Press PTT to save.
5. Confirm channel 17 and code 1 on the display.

T461 guidance uses Motorola's T4xx-family manual. Verify the menu labels on your unit.

OPERATIONS SECTION

Before the Beaver Brigade rolls out

A two-way check from every vehicle

- Start on primary channel 16 with CTCSS 67.0 Hz and narrow bandwidth.
- Test every vehicle on channel 16, then switch together to channel 17 and test again.
- Lead calls each vehicle. Each driver replies so both directions get checked.
- Return everyone to channel 16. Keep antennas upright and switch only as a group.

If someone cannot hear

- Check volume and the selected direct channel.
- Turn receive filtering OFF or use Monitor. If audio returns, the tone settings differ.
- If audio is quiet or distorted, check bandwidth and microphone technique.
- GMRS: say your call sign at the end of a series of transmissions and at least every 15 minutes during a series.

Repeater reference

For MXT275, MXT575 and GM-15 Pro owners

(GXT1000 and Motorola T4xx use direct channels and cannot use repeaters)

Repeater basics

A shared relay on higher ground

Everyone must select the same actual repeater and be within its coverage.

Radio receives on 462 MHz and transmits 5 MHz higher on 467 MHz.

A regular channel 16 and a repeater channel 16 are different operating modes.

The owner's settings control access

Use an authorized repeater and its required transmit tone or DCS code.

A tone is common, but not mandatory on every repeater.

Receive tone is optional unless needed for your setup. It must match the repeater's output tone, which can differ from its input tone.

GMRS license required. Use the repeater owner's frequency pair, tones and bandwidth.

Repeater setup on compatible radios

Midland MXT275 / MXT575

Enable repeater channels in menu rP.

Select the owner's channel with RP visible.

Set tC to the owner's input tone. Use the actual tone chart to choose its code number.

Set rC to OFF initially, or the known output tone. Confirm changes with LOCK.

Baofeng GM-15 Pro

Choose the factory repeater channel for the owner's frequency pair.

Set TX CTCSS or TX DCS to the required input value.

Start with receive filtering OFF. Add RX tone only after confirming the output tone.

Use the owner's bandwidth setting. Confirm the frequency pair and test both directions.